

Future Architecture — Evolution Paths

Future-plans — evolution paths, triggers, cost trajectories

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Future-plans — evolution paths, triggers, cost trajectories

Scope: The architecture in `helm/ + infrastructure/` is shipped at ONE operating point — POC scale, `cells.count=1`, single master AZ, cold DR via Velero. This directory documents how that operating point evolves as customer scale, SLA pressure, compliance demand, or industry shift moves the architecture's centre of gravity.

How to use this directory: this README is the umbrella + the time-and-business axes. The individual docs below are single evolution paths with their own depth. Read the README for the *map*; drill into a spoke when a trigger fires.

The architecture stays invariant inside one horizon — only `values.yaml` configuration moves. Crossing a horizon is structural, not parameter.

1. Architecture evolution map

Five horizons. Each row names the business shape, the architecture shape, and the trigger that moves from the previous horizon. The pairs of horizons are bracketed, not precise — "T+3y" means "the phase where the T+3y trigger conditions typically appear for Mittelstand-scale B2B SaaS"; some customers reach it in 18 months, some never do.

Horizon	Typical business shape	Architecture shape	Trigger to move from previous
T0 (today)	POC / early-stage; ~5–10	Single master AZ, cold DR via Velero, dual-	—

Horizon	Typical business shape	Architecture shape	Trigger to move from previous
	tenants total; 1 cell	cadence backup (5-min operational + 4-h DR), <code>cells.count=1</code>	
T+1y	Early customers; ~10–50 tenants spread over ~3 cells; cohort mix plus 1–2 enterprise-dedicated pods	Same shape; scale <code>cells.count</code> linearly via ADR-02 Mode 2 (empty-pod fill); Karpenter handles stateless tier elasticity	Aggregate cell utilisation > 70% sustained two weeks
T+3y	Growth; ~50–100 tenants over ~10–30 cells; some hot tenants need their own cell; mixed cohort + enterprise	EITHER same single-master-AZ shape scaled linearly OR substrate upgrade to TiKV multi-AZ HA (per <code>tikv-upgrade-path.md</code>) if SLA tightens	Customer SLA tightens RTO below ~25 min AZ failure OR cost-of-cold-DR exceeds cost of distributed-K-V running cost
T+5y	Mature SaaS; cross-region customer base; multi-region active-active may become product feature	Multi-region active-active — TiCDC async between regional TiKV clusters OR Path A geo-close 5-node Raft (see <code>tikv-upgrade-path.md</code> § 4)	Cross-continent customer base + sub-50 ms read latency demand OR multi-region write-everywhere becomes the product surface
T+10y	Market maturity; substrate-level questions emerge	LevelDB EOL question; serverless distributed K-V (DSQL / Spanner-equivalent SaaS) maturity question; AI-augmented placement / vector retrieval as adjacent systems	Industry inflection — when a serverless-DB offering's SLA + cost + portability collapse the current architecture's complexity, OR LevelDB itself gets deprecated by upstream maintainers

Two structural inflection points are visible in this map:

- **T+1y → T+3y:** the *option* to upgrade the storage substrate (LevelDB → TiKV). The current architecture stays valid linearly through T+3y; the upgrade is a customer-SLA-driven choice, not a forced step.
- **T+5y → T+10y:** the *substrate question itself* — does in-cluster distributed K-V remain the right primitive, or does the market move to serverless databases that obviate the question?

Inside a horizon, configuration moves (`cells.count`, `backup.dr.cadence_hours`, `master_az`, etc.). Crossing a horizon, the architecture shape changes.

2. Trigger conditions — what moves us between horizons?

Five axes; ANY ONE tripping is enough to start the conversation. Documented per axis so "are we already at the trigger?" is answerable from operational telemetry, not subjective judgement.

2a. Tenant-count / capacity axis

Phase boundary	Trigger metric	Threshold	Source of truth
T0 → T+1y	Aggregate cluster utilisation	> 70% sustained two weeks	per-tenant-ops Grafana dashboard
T+1y → T+3y	Per-cell utilisation	> 85% sustained four weeks (Mode 1 per ADR-02)	capacity-headroom dashboard
T+3y → T+5y	Single largest tenant size	> 50% of one pod, AND tenant-cell count > 50	per-tenant-ops dashboard
T+5y → T+10y	Substrate operational burden	Operator-survey "we're managing the database, not the product"	qualitative; quarterly review

2b. SLA / RTO / RPO axis

Trigger	Architecture implication
Customer SLA tightens RTO below the ~25 min AZ-failure budget	TiKV multi-AZ HA (RTO ~0); see <code>tikv-upgrade-path.md</code> § 2
Customer SLA demands sub-15-min cross-region RTO	Multi-region active-active; CAP conversation; <code>tikv-upgrade-path.md</code> § 4
RPO target tightens below 5-min cadence floor	WAL shipping or distributed consensus required; substrate change
Granular per-tenant point-in-time restore becomes a product feature	Velero FSB path; see <code>restic-fsb-patch.md</code>

2c. Compliance axis

Trigger	Architecture implication
Regional data residency commitment (e.g., GDPR Schrems II sustained ruling)	Per-region cell pool + tenant-to-region pinning at the placement-table layer
SOC 2 Type II / ISO 27001 audit commitment	Backup-verification automation (currently in ADR-04 "out of POC"); EBS Fast Snapshot Restore (FSR) enabled for evidence-quality restore drills

Trigger	Architecture implication
HIPAA / financial-tier compliance	Tier 1 isolation per ADR-04 (dedicated AWS account per customer); separate KMS hierarchy
Air-gap requirement	Restic FSB path with offline-restore-from-USB (see <code>restic-fsb-patch.md</code> § "When Restic would be the right call")

2d. Cost-ceiling axis

Phase	Total monthly cost (illustrative; see § 3 line-by-line)	Lever revisited
T0	~\$2,700	none — within Mittelstand POC budget
T+1y	~\$5,000–8,000	NAT Gateway consolidation; 1-year Savings Plans for stateful baseline
T+3y (same shape)	~\$15,000–25,000	Karpenter spot diversification; multi-tenancy consolidation within cohort cells
T+3y (TiKV substrate)	~\$20,000–30,000	Higher compute (3× Raft replicas) offset by lower backup pipeline cost; 3-year Savings Plans
T+5y (multi-region active-active)	~\$50,000–100,000	Cross-region traffic is the dominant line item; depends on TiCDC sync volume

2e. Industry-shift axis

Trend	When relevant	Architecture implication
Serverless distributed K-V matures into production-grade SLA + cost	T+5y if Aurora DSQL / Spanner-equivalent become viable for B2B SaaS data	Substrate change; remove our own cluster ops
Edge-native databases for global latency (Planetscale, Neon, Turso class)	T+5y if customer demand for sub-100 ms global writes emerges	Application-level rewrite; different cost model
AI-augmented retrieval (vector DBs, feature stores)	T+3y if AI features become a product surface	Adjacent system (Pinecone / pg_vector / OpenSearch); not a LevelDB replacement
Kubernetes maturity + operator-pattern dominance	already true; reinforces T+3y TiKV path	The migration risk is bounded; operator-pattern tools are battle-tested

3. Cost trajectory — broken down per horizon

All figures derive from formulas in [docs/operations/cost-estimate-methodology.md](#) — operator substitutes actual customer scale variables to land the real number. The point of this section is the *shape* of cost evolution, not the precise number. AWS pricing as of 2026; operator verifies against current pricing before committing.

3a. T0 — POC scale (`cells.count=1`)

Line item	Cost (USD/mo)	Formula
EKS control plane	\$73	flat, 1 cluster
Stateful nodes (master AZ only)	\$500	$r6id.2xlarge \times 1 \text{ pod} \times 730 \text{ h} + \text{on-demand overhead}$
Stateless nodes (api + envoy + system)	\$800	Karpenter mixed 70% spot / 30% on-demand, ~3 nodes equivalent
EBS gp3 storage (master AZ only)	\$200	$2 \text{ TB} \times \$0.08/\text{GB}$; standby AZ at \$0 (no PVCs until rotation)
3× NAT Gateways	\$99	$3 \times \$33$ (per-AZ; preserves warm-standby rotation readiness per ADR-10 §6)
EBS Snapshot (5-min operational + 4-h DR cadence, 30-day retention)	\$300	depends on churn (10–20%/day) → 1.5–3 TB at retention
Cross-region snapshot copy (Glacier Instant Retrieval)	\$150	$\text{replicated changed-data} \times \$0.004/\text{GB} + \text{transfer}$
S3 (Velero metadata + replication)	\$50	small footprint (manifests + state)
Observability (Grafana Cloud Pro)	\$500	mid-tier subscription; AMP/AMG alternative ~\$700
Subtotal	~\$2,700	Mittelstand POC budget

3b. T+1y — Mid-stage (~10 customers, `cells.count=3`)

Line item	Cost (USD/mo)	Δ vs T0
EKS control plane	\$73	unchanged (one cluster covers all cells)
Stateful nodes	\$1,500	$\times 3$ cells (linear)
Stateless nodes	\$1,000	scales with API rate (~1.25×)
EBS gp3 storage	\$600	$\times 3$ cells (linear)
3× NAT Gateways	\$99	unchanged (one VPC, three AZs)
EBS Snapshot	\$900	$\times 3$ cells (linear)
	\$450	$\times 3$ cells (linear)

Line item	Cost (USD/ mo)	Δ vs T0
Cross-region snapshot copy		
S3 metadata	\$100	more namespaces, more snapshot manifests
Observability	\$500	unchanged (custom-metric growth modest)
Subtotal	~\$5,200	Linear scaling kicks in; near-linear in <code>cells.count</code>

3c. T+3y, Path A — Same shape, scaled (`cells.count=20`)

Line item	Cost (USD/ mo)	Note
Stateful nodes	\$10,000	$\times 20$ cells (linear)
EBS storage	\$4,000	$\times 20$ cells (linear)
Snapshots + cross-region	\$9,000	$\times 20$ cells (linear)
Stateless + NAT + obs + EKS	\$2,500	non-linear modest growth
Subtotal	~\$25,500	Linear path; lowest-disruption choice

3d. T+3y, Path B — TiKV multi-AZ HA substrate

Line item	Cost (USD/ mo)	Δ vs Path A
TiKV nodes (3-replica across 3 AZs)	\$14,000	$\sim 3\times$ compute vs single-master-AZ; node count grows
EBS storage	\$4,000	same per-replica; Raft amortises some compaction differently
Snapshot pipeline	\$2,000	smaller — TiKV's own snapshot/Raft log replaces high-cadence Velero pipeline
Cross-region copy	\$4,000	smaller — less data churn in dedicated DR snapshots
Other (PD, NAT, obs, EKS)	\$4,000	TiKV adds PD overhead; multi-AZ tripled NAT data-processed
Subtotal	~\$28,000	$\sim 10\%$ premium over Path A; buys RTO ~ 0 + active-AZ across all 3 AZs

The choice between Path A and Path B at T+3y is SLA-driven: if RTO ~ 25 min still fits the customer's tolerance, Path A is correct (cheaper, simpler). If RTO target tightens, Path B's $\sim 10\%$ cost premium buys structural correctness.

3e. T+5y — Multi-region active-active (if customer SLA forces it)

This is the inflection point where architecture cost stops being linear:

Line item	Cost (USD/ mo)	Driver
Two-region TiKV clusters	\$50,000	~2× of Path B for compute + storage
Cross-region traffic (TiCDC bidirectional sync)	\$5,000–20,000	depends on write volume; dominant variable
Operational complexity premium	\$5,000–10,000	multi-region SRE expertise; deeper observability
Subtotal	~\$60,000–80,000	Per ~500 customers; varies with write volume

3f. T+10y — Substrate uncertain

Cost shape depends on which substrate dominates the market:

- **If serverless distributed K-V matures:** cost could *drop* dramatically (no cluster operations; pay-per-usage). Architecture becomes a thin layer above a managed primitive.
- **If edge-native databases win:** cost shape changes shape entirely (latency-billed per region, not capacity-billed per cluster).
- **If status quo persists** (TiKV / Cassandra / FoundationDB remain the answer): cost scales with the patterns above.

Hard to forecast precisely; depends on industry trajectory. Reserved for "we'll know when we get there" reasoning.

4. Future-plan documents — index

Each doc is one evolution path with its own depth + trade-off analysis + migration cost.

Doc	Evolution type	Trigger that activates it
<u>restic-fsb-patch.md</u>	Tactical — same architecture, different backup transport (Velero CSI → FSB / Kopia / Restic)	Customer's RTO tolerance ≥ 12 h OR granular per-tenant file restore becomes product feature OR air-gap requirement
<u>lvm-init-patch.md</u>	Tactical — same architecture, literal-LVM storage stack (Path γ multi-PVC → Path α LVM init container)	Existing legacy on-prem LevelDB host runs LVM and migration target should match mental model OR customer policy mandates literal Linux storage stack OR in-host thin snapshot independent of CSI is required

Doc	Evolution type	Trigger that activates it
tikv-upgrade-path.md	Strategic — substrate change (LevelDB → TiKV distributed K-V)	RTO ~25 min AZ failure not acceptable OR sub-5-min RPO required OR multi-region active-active becomes product requirement
tikv-upgrade-path/migration-runbook.md	Operational — phase-by-phase playbook for the LevelDB → TiKV cutover (companion to the strategic doc above)	After Stage 3 confirms the trigger fired AND customer committed to weeks of app engineering

More docs land here as the architecture evolves. Each starts as a [planning] doc, gets reviewed, then either ships or stays as reading material until its trigger fires.

5. Industry trend context — where the space is heading

A brief survey of where K8s-stateful + distributed-K-V is in 2026 and where the current architecture sits relative to that trajectory.

Distributed-K-V on K8s is maturing. TiKV, Cassandra, FoundationDB, ScyllaDB all run on K8s in production at scale. Operator-pattern deployments (PD, etcd-operator, TiKV operator) are now the canonical shape. The current architecture's single-LevelDB-per-pod is the *not-yet-distributed* corner of this design space; the TiKV path documents the migration if the customer's SLA pressure outgrows what LevelDB physics can deliver.

Serverless databases are emerging. AWS Aurora DSQL (announced 2024), AlloyDB Omni (GCP), Cloudflare D1, Turso, Neon. If these mature with the operational characteristics customers expect (SLA + cost + portability + data-residency), they could collapse the current architecture's complexity into a managed primitive. The T+5y horizon question is whether this happens fast enough to matter.

Edge-native databases for latency-sensitive workloads. PlanetScale, Neon, Turso class. Cost model is materially different (read-replica-per-region vs cluster-per-region). Right answer for global SaaS where read latency dominates; not yet a fit for write-heavy B2B SaaS with strong-consistency demand.

AI-era retrieval primitives. Vector retrieval (Pinecone, pg_vector, OpenSearch), feature stores (Feast, Tecton). These are adjacent systems, not LevelDB replacements; integration happens at the application layer. T+3y horizon if AI features become a product surface; doesn't affect the storage substrate decision.

Kubernetes operator-pattern maturity. Battle-tested operators (etcd, TiKV, Cassandra, PostgreSQL via Crunchy / CloudNativePG) reduce the operational risk of running a distributed database on K8s. This *de-risks* the T+3y TiKV path — the bet that the operator handles the hard parts is much safer in 2026 than it was in 2020.

Where the current architecture sits. Deliberately conservative — optimised for "Mittelstand SaaS where the storage primitive is the application's existing LevelDB". This is the right call at T0; the future-plan docs above name the conditions under which it should change.

6. Caveats

- **All cost numbers are illustrative.** Real costs depend on customer-specific traffic shape, retention policy, KMS request volume, observability custom-metric count, AWS pricing in effect at the time. Operator must substitute actuals via [docs/operations/cost-estimate-methodology.md](#).
- **All triggers are minimums.** Hitting one trigger starts the architecture-conversation; it doesn't auto-mandate the change. Many triggers can fire harmlessly if the customer accepts the trade-off.
- **All timelines are bracketed, not precise.** "T+3y" doesn't mean "exactly 36 months from now"; it means "the phase where the T+3y trigger conditions typically appear for Mittelstand-scale B2B SaaS". Some customers reach T+3y in 18 months; some never do.
- **All alternatives are situational.** TiKV is the canonical reference but not the only answer (see [tikv-upgrade-path.md](#) § 7 for the four alternatives: CockroachDB, ScyllaDB, FoundationDB, DynamoDB Global Tables). The right substrate depends on which CAP corner + operational model + ecosystem fits the customer.
- **All horizons are reviewable.** The right cadence to re-read this README is quarterly — once a quarter, look at where the operational telemetry actually is against the trigger thresholds in § 2.